hw9

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Chapter 18

Question 12

```
p_unvax <- 0.00369
p_vax <- 0.001
population <- 100000

odds_unvax <- p_unvax / (1 - p_unvax)
odds_vax <- p_vax / (1 - p_vax)

odds_ratio <- odds_unvax / odds_vax
print(odds_ratio)
```

```
## [1] 3.699963
```

Unvaccinated people have 3.7 higher odds of getting the disease, so the protection rate is 73%.

```
(p_unvax *population - p_vax*population) / (p_unvax *population)
```

```
## [1] 0.7289973
```

```
p_unvax2 <- 0.48052
p_vax2 <- 0.2

odds_unvax2 <- p_unvax2 / (1 - p_unvax2)
odds_vax2 <- p_vax2 / (1 - p_vax2)

odds_ratio2 <- odds_unvax2 / odds_vax2
print(odds_ratio2)
```

```
## [1] 3.700008
```

```
population <- 100000

odds_unvax <- p_unvax2 / (1 - p_unvax2)
odds_vax <- p_vax2 / (1 - p_vax2)
```

```r
odds_ratio <- odds_unvax / odds_vax

cases_unvax <- p_unvax2 * population
cases_vax <- p_vax2 * population

cases_prevented <- cases_unvax - cases_vax
protection_rate <- cases_prevented / cases_unvax
print(protection_rate)
```

```
## [1] 0.5837842
```

The protection rate has dropped to 58%.

Chapter 20

Question 11

part a

```r
fail_model <- glm(Failure ~ Temperature, family = binomial, data=ex2011)
summary(fail_model)
```

```
##
## Call:
## glm(formula = Failure ~ Temperature, family = binomial, data = ex2011)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) 10.87535    5.70291   1.907   0.0565 .
## Temperature -0.17132    0.08344  -2.053   0.0400 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 28.975  on 23  degrees of freedom
## Residual deviance: 23.030  on 22  degrees of freedom
## AIC: 27.03
##
## Number of Fisher Scoring iterations: 4
```

part b

```r
coef(fail_model)
```

```
## (Intercept) Temperature
##  10.8753491  -0.1713205
```

The model shows a coefficient of temperature that is different from , at -0.17132.

part e

```r
prob_31 <- predict(fail_model, newdata = data.frame(Temperature = 31), type = 'response')
print(prob_31)
```

```
##         1
## 0.9961828
```

part f

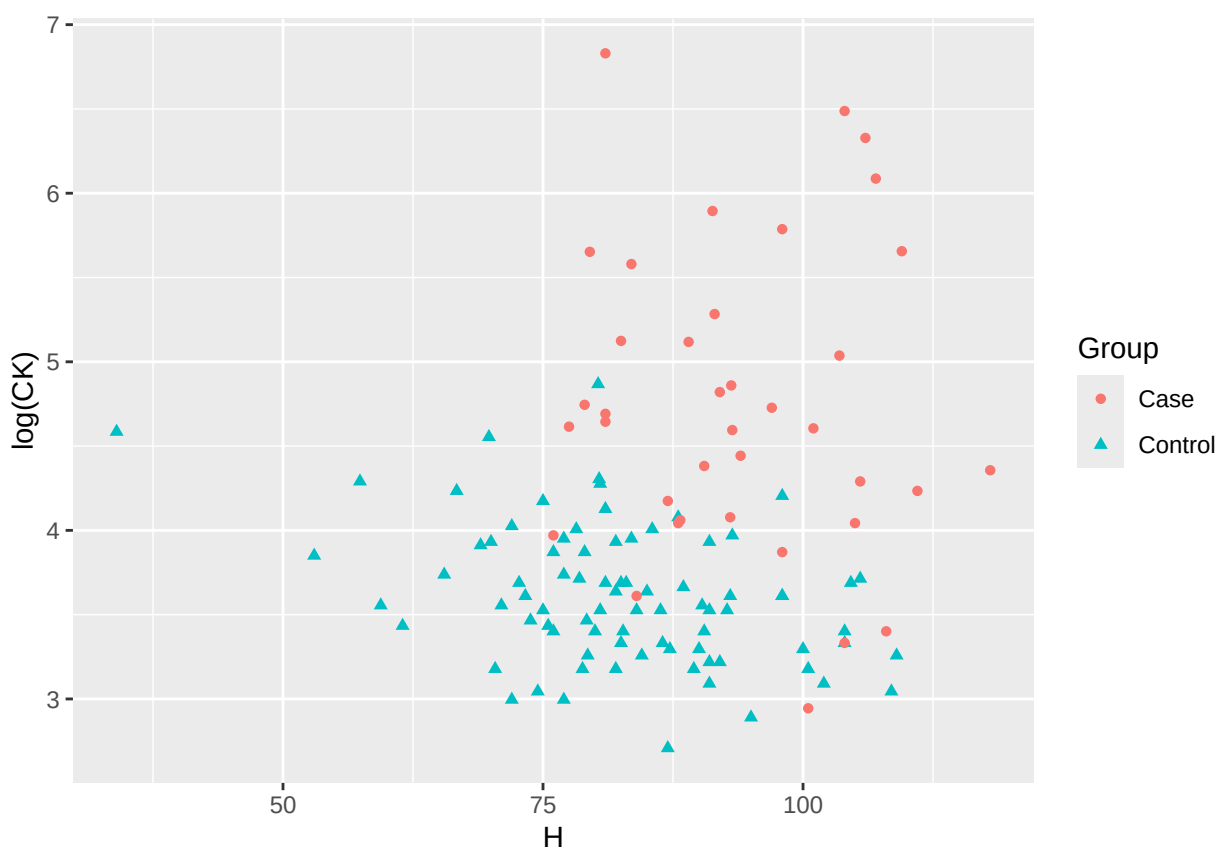
The answer to part e must be treated cautiously because it involves extrapolation, making a prediction outside the range of the available explanatory variable values. In this case, the logistic regression model was fitted using temperature data from 53 degree F to 81 degrees F, so predicting the O-ring failure at 31 degrees F, which is outside this range.

Extrapolation is risky because the relationship shown in the data might not be present outside of it, especially when 31 degrees is below freezing, which is quite important for temperature failure probabilities.

Question 12

part a

```r
library(ggplot2)
ex2012 |> ggplot(aes(x=H, y=log(CK), shape = Group, color = Group)) + geom_point()
```



It appears that enzymes might be a useful predictors of whether or not a women is a carrier.

part c

```r
disease_mod <- glm(Group ~ log(CK) + H, data = ex2012, family = "binomial")
summary(disease_mod)
```

```
##
## Call:
## glm(formula = Group ~ log(CK) + H, family = "binomial", data = ex2012)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) 28.91340    5.80017   4.985 6.20e-07 ***
## log(CK)     -4.02043    0.82910  -4.849 1.24e-06 ***
## H           -0.13652    0.03654  -3.736 0.000187 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 149.840  on 119  degrees of freedom
## Residual deviance:  61.992  on 117  degrees of freedom
## AIC: 67.992
##
## Number of Fisher Scoring iterations: 7
```

When increasing H by 1, you increase the log(odds) of being a DMD disease carrier by 0.14, which multiplies odds by e^0.14 = 1.14.

part e

```r
predict(disease_mod, data.frame(CK = c(300,80), H = c(100, 85))) |> diff() |> exp()
```

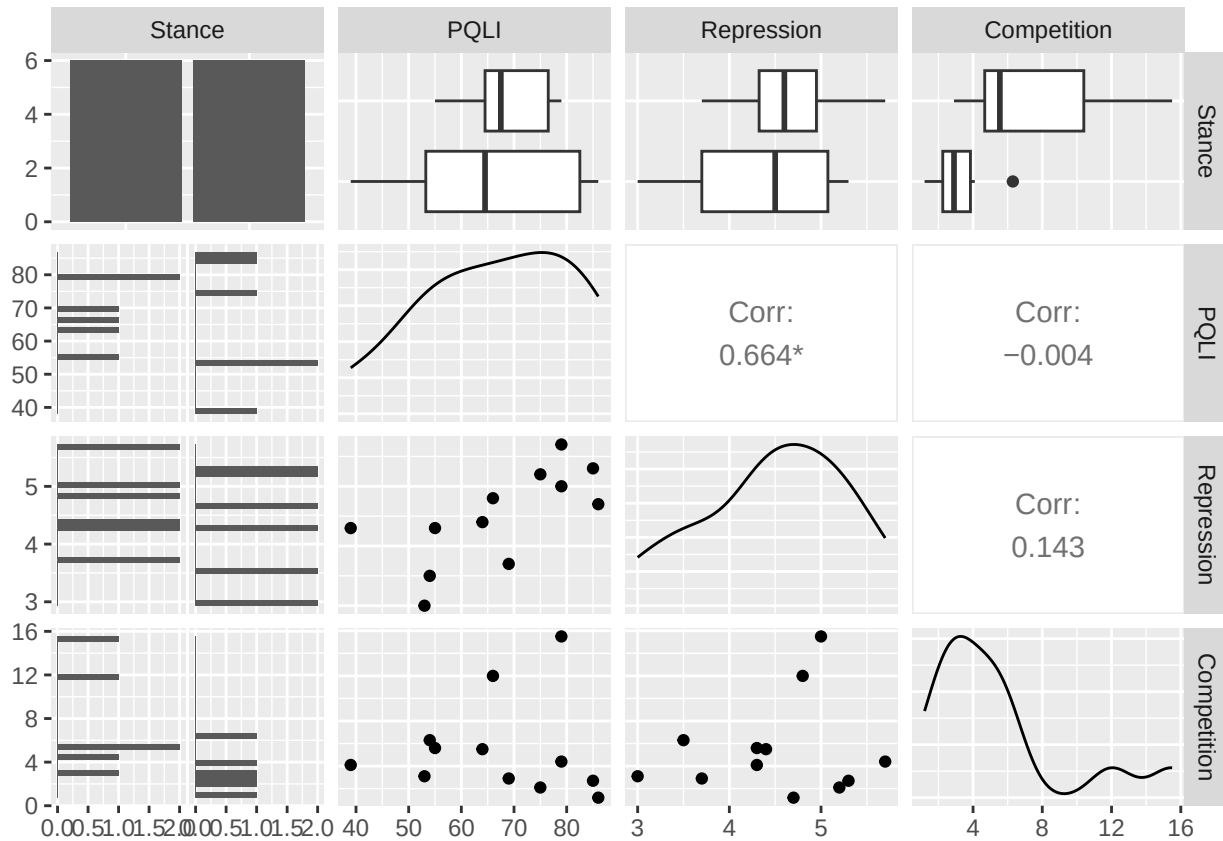
```
##        2
## 1574.675
```

Question 17

```r
ex2017 |> select(-Country) |> GGally::ggpairs()
```

```
## Registered S3 method overwritten by 'GGally':
##   method from
##   +.gg   ggplot2
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```




```r
glm(Stance ~ PQLI + Repression + Competition, data=ex2017, family='binomial') |> summary()
```

```
##
## Call:
## glm(formula = Stance ~ PQLI + Repression + Competition, family = "binomial",
##     data = ex2017)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  9.54627    7.31063   1.306    0.192
## PQLI        -0.08110    0.10927  -0.742    0.458
## Repression   0.06443    1.48133   0.043    0.965
## Competition -1.02979    0.68037  -1.514    0.130
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 16.636  on 11  degrees of freedom
## Residual deviance: 10.211  on  8  degrees of freedom
## AIC: 18.211
##
## Number of Fisher Scoring iterations: 7
```

While the pairs graph suggest that competition might be influential factor for the Catholic church's stance, none of the variables are significant, so no conclusions can be made.

Question 20

```
self_esteem <- ex1223 |> mutate(esteem = (Esteem1 == 1))
self_esteem |> mutate(logIncome = log(Income2005)) |>
select(Educ, logIncome, AFQT, Gender, esteem) |>
GGally::ggpairs()
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
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## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```

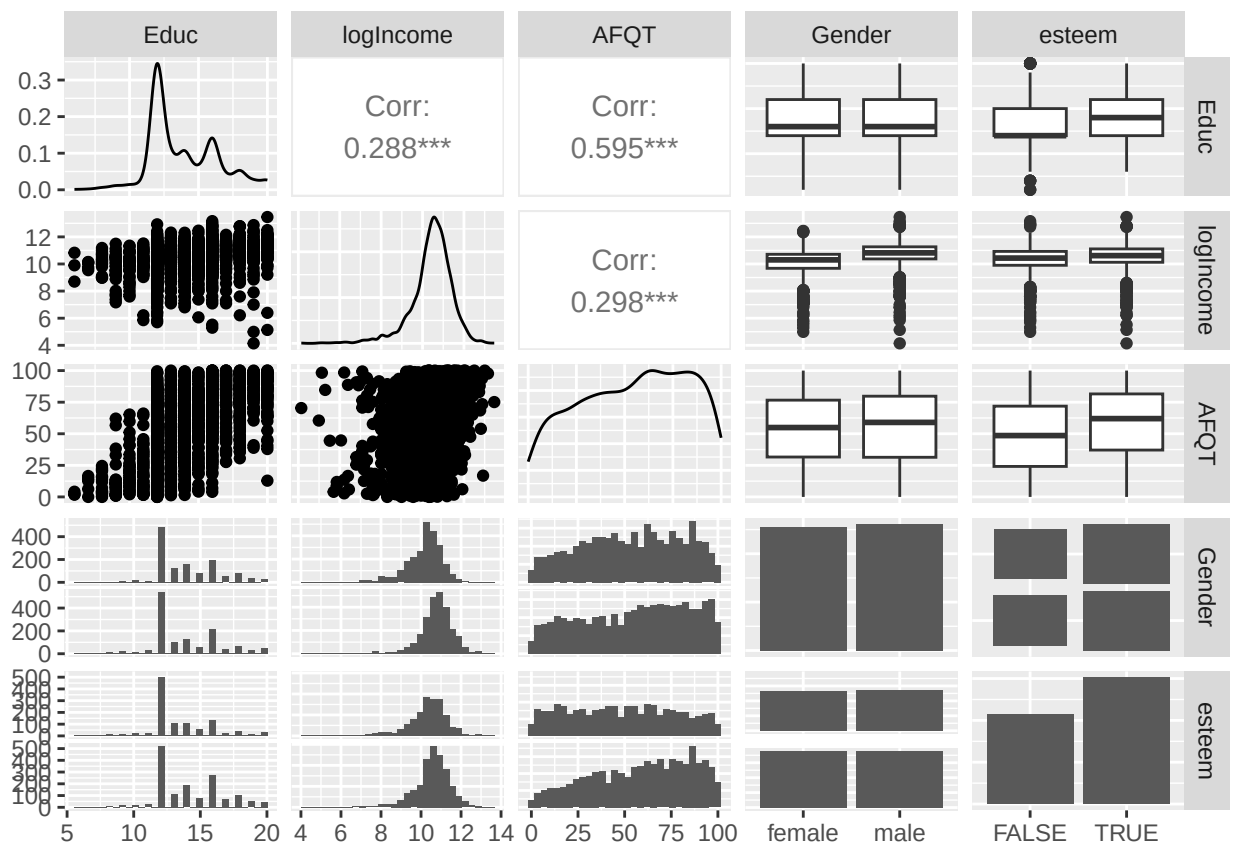


```
self_e <- glm(esteem ~ log(Income2005) + AFQT + Educ + Gender, data=self_esteem, family='binomial')
summary(self_e)
```

```
##
## Call:
## glm(formula = esteem ~ log(Income2005) + AFQT + Educ + Gender,
##     family = "binomial", data = self_esteem)
##
## Coefficients:
```

```
##                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)     -2.773457   0.485833  -5.709 1.14e-08 ***
## log(Income2005)  0.165970   0.047114   3.523 0.000427 ***
## AFQT             0.007603   0.001842   4.128 3.65e-05 ***
## Educ             0.076135   0.021045   3.618 0.000297 ***
## Gendermale      -0.146214   0.087097  -1.679 0.093201 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 3508.3  on 2583  degrees of freedom
## Residual deviance: 3395.5  on 2579  degrees of freedom
## AIC: 3405.5
##
## Number of Fisher Scoring iterations: 4
```

AFQT has the most significant p-value, but educ and income also are significant, while gender does not have a significant effect.